Acids and Bases

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Contents

1 Acid-Base Basics

Core Concept 1.1: Acids

Acids are substances that increase the $[H^+]$ of a solution.

(shorthand) $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

(more accurately) $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

Core Concept 1.2: Bases

Bases are substances that decrease the $[H^+]$ of a solution (usually through the addition of OH^-, however there are exceptions).

(shorthand) $BOH(aq) + H^+(aq) \rightleftharpoons B^+(aq) + H_2O(aq)$

(more accurately) $BOH(aq) + H_3O^+(aq) \rightleftharpoons B^+(aq) + 2\,H_2O(l)$

Core Concept 1.3: Brønsted-Lowry Theory

Brønsted-Lowry Theory defines acids as H^+ donors and bases as H^+ acceptors (not just OH^- donors).

As covered in Equilibrium, acid-base reactions also have equilibriums (*denoted as* $K_{a/b}$) specific to each acid/base.

Core Concept 1.4: Strong/Weak Acids and Bases

Acid-base reactions with high $K_{a/b}$ are categorized as **strong acids/bases** ($K_{a/b} >> 1$) and those with low $K_{a/b}$ as **weak acids/bases** ($K_{a/b} << 1$). Strong acids and bases also tend to conduct electricity better because of more dissociated ions.

Core Concept 1.5: Indicators

Indicators (In) are weak acids or bases that change color at some $[H^+]$ in a solution. The color change occurs because the corresponding acid (HIn) and base (In^-) forms of the indicators possess different colors.

$HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq)$

thus K_{In} is defined as

$$K_{In} = \frac{[H^+][In^-]}{[HIn]}$$

Here are a two examples of indicators that are commonly used in the lab setting.

Anthocyanin: $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

Anthoxanthin: $HB(aq) \rightleftharpoons H^+(aq) + B^-(aq)$

Anthocyanin shows either a red or blue hue based on the comparative level of its acid form to its corresponding base. Anthoxanthin, on the other hand, is either clear or yellow. Through Le Chatelier's principle, we can manipulate the color of the indicator solution by adding a strong acid or base to raise or lower $[H^+]$, respectively.

2 Acid-Base Stoichiometry

Core Concept 2.1: Amphiproticity

Water is **amphiprotic**, meaning that it can both accept or donate an H^+.

(H^+ Acceptor) $HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$

(H^+ Donor) $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

Water also maintains equilibrium with the autoionized form of itself. Heat is a reactant in this autoionization reaction, thus an increase or decrease in temperature will shift the reaction right and left, respectively.

$H_2O(l) + H_2O(l) + \text{heat} \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

Therefore the equlibrium of $H_2O(l)$ is represented as

$$K_W = [H3O^+][OH^-] = [H^+][OH^-]$$

Core Concept 2.2: pH

pH is a quantitative measure of how acidic (*low pH*) or basic (*high pH*) a solution is. Mathematically, pH is defined as

$$pH = -\log_{10}[H^+]$$

Water temperature (°C)	K_w	$[H^+]$	pH
0	$10^{-14.94}$	$10^{-7.47}$	7.47
5	$10^{-14.73}$	$10^{-7.36}$	7.36
10	$10^{-14.53}$	$10^{-7.26}$	7.26
15	$10^{-14.35}$	$10^{-7.18}$	7.18
20	$10^{-14.17}$	$10^{-7.08}$	7.08
25	$10^{-14.00}$	$10^{-7.00}$	7.00
30	$10^{-13.83}$	$10^{-6.92}$	6.92
35	$10^{-13.68}$	$10^{-6.84}$	6.84
40	$10^{-13.53}$	$10^{-6.76}$	6.76

Table 1: Water pH across different temperatures

A common misconception people often assume is that the pH of 7 corresponds to a neutral solution of water ($[H^+] = [OH^-]$). From Table 1 however, we see that **the pH of 7 being neutral only applies to a temperature of 25°C**. At higher temperatures, the equilibrium will shift right to produce more H_3O^+, effectively making a lower pH value as the new neutral base point.

From a molecular perspective, higher temperatures increases autoionization rates because the higher kinetic energy of the water molecules makes it statistically more likely to collide/react with one another in order to form H_3O^+ and OH^-, thus decreasing the pH of the solution.

Let's take a moment to look at stoichiometric calculations for acids and bases.

Core Concept 2.3: Strong Acid/Base Stoichiometry

For **strong acids and bases**, we rely on simple stoichiometry to calculate $[H^+]$. We'll use HCl as an example.

$HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$

The equilibrium for HCl is represented as

$$K_a = \frac{[H_3O^+][Cl^-]}{[HCl]} >> 1$$

The $K_{a/b}$ for strong acids and bases are generally very large values. The approximate K_a for HCl is 10^7. Thus when dealing with strong acids and bases, we can assume the initial [HCl] to be equal to the final $[H3O^+]$ with a negligible margin of error.

$$[HCl] = [H_3O^+] = [H^+]$$

Subsequently, you can calculate pH using the formala $pH = -\log_{10}[H^+]$

Core Concept 2.4: Weak Acid/Base Stoichiometry

For **weak acids and bases**, we rely on more complex stoichiometry to calculate $[H^+]$. We'll use acetic acid (CH_3COOH) as an example.

$CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$

The equilibrium for acetic acid is represented as

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} << 1$$

The $K_{a/b}$ for weak acids and bases are generally very small values. The approximate K_a for acetic acids is 1.8×10^{-5}. Thus when representing the K_a of acetic acid, we can derive the equation

$$K_a = 1.8 \times 10^{-5} = \frac{[H_3O^+]_{final}[CH_3COO^-]_{final}}{[CH_3COOH]_{initial} - [H_3O^+]_{final}} \approx \frac{[H_3O^+]_{final}[CH_3COO^-]_{final}}{[CH_3COOH]_{initial}}$$

Here, since a mole of CH_3COOH is consumed every mole of H_3O^+ or CH_3COO^- is formed, $[CH_3COOH]_{final}$ must be equal to $[CH_3COOH]_{initial} - [H_3O^+]_{final}$. However, since K_a is extremely small, we approximate $[CH_3COOH]_{final}$ as just $[CH_3COOH]_{initial}$ because $[H_3O^+]_{final}$ value is negligible in comparison to $[CH_3COOH]_{initial}$.

The smaller the $K_{a/b}$ of a weak acid or base is, the more accurate the stoichiometric calculation will be.

Core Concept 2.5: Conjugate Acid-Base Pairs

The corresponding base that is formed when an acid donates a H^+ is called the **conjugate base**. Likewise, the corresponding acid that forms when a base accepts a H^+ is called the **conjugate acid**.

For example, the conjugate acid for NH_3 is NH_4^+, and the conjugate base for HNO_3 is NO_3^-. These are represented as

$NH_3(aq) + H^+(aq) \rightleftharpoons NH_4^+(aq)$

$HNO_3(aq) \rightleftharpoons H^+(aq) + NO_3^-(aq)$

Because the chemical reaction for an acid is the same as that of the conjugate base (*just reversed*), the K_a for an acid is the inverse of the K_b for its conjugate base. As such, the stronger the acid/base, the weaker the corresponding conjugate base/acid will be.

Core Concept 2.6: Conjugate Acid-Base Pair Equilibriums and K_w

The product of the equilibrium constants of an acid K_a and its conjugate base K_b (*or base and its conjugate acid*) is equal to the equilibrium constant of the autoionization of water K_w. We can derive this relationship with the following algebraic manipulations using the formulas for the shell reactions below.

(Acid) $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

(Conjugate Base) $A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq)$

Thus the equilibrium constants are $K_a = \frac{[H_3O^+][A^-]}{[HA]}$, $K_b = \frac{[HA][OH^-]}{[A^-]}$. When we rewrite K_b as

$$K_b = \frac{[HA][OH^-]}{[A^-]} \times \frac{[H_3O^+]}{[H_3O^+]} = K_w/K_a \Rightarrow K_w = K_aK_b$$

This relationship is true for all conjugate acid-base pairs.

3 Titration and Buffers

Core Concept 3.1: Titration

Titration is the process of adding an acid or base to bring it about to a level where occurs a change at a previously known pH (via indicators). This laboratory technique **allows chemists to determine the pH of a solution** by observing the amount of base or acid it takes to reach the target pH level.

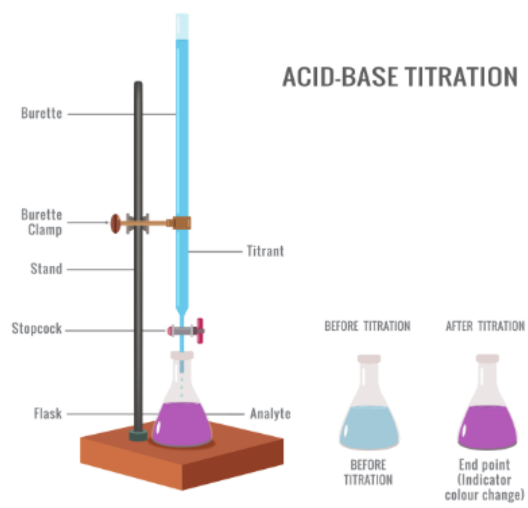

Figure 1: Basic titration set up

We call the solution that is being titrated the **analyte** and the solution that is doing the titrating the **titrant**.

Core Concept 3.2: Strong Acid and Strong Base Titration

An example of a **strong acid** (HCl) being titrated with a **strong base** (NaOH) is represented by the chemical equation $HCl(aq) + NaOH(aq) \rightleftharpoons NaCl(aq) + H_2O(l)$. Because of their high $K_{a/b}$ values, we can simplify this equation to the following.

$$H^+(aq) + OH^-(aq) \rightleftharpoons H_2O(l)$$

To neutralize a solution of an unknown [HCl], we would need to add equal amounts of NaOH. Thus, based on the amount of NaOH we add to the solution in order to reach this neutral point, we can calculate the original [HCl].

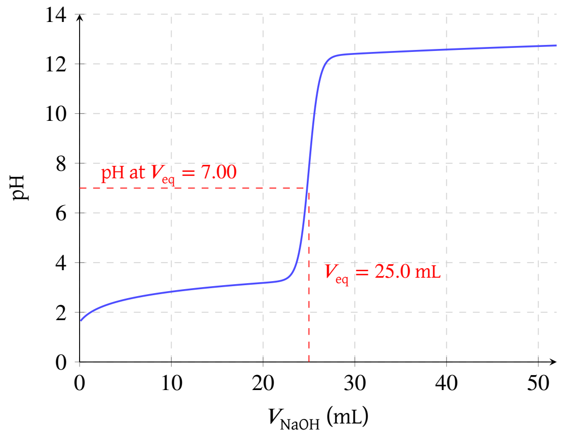

Figure 2: Strong acid-base titration curve

Figure 2 is a graphical representation of the analyte's pH as more titrant is added. The point at which the moles of titrant added is stoichiometrically equal to the amount of the analyte originally present is called the **equivalence point**. For the case of a strong acid with a strong base, this point is at a neutral pH of 7 at 25°C.

Core Concept 3.3: Weak Acid/Base and Strong Base/Acid Titration

The titration curve of a **weak acid** (CH_3COOH) being titrated by a **strong base** (NaOH) is shown below.

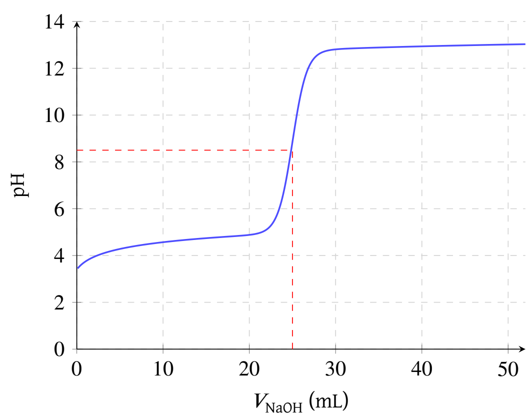

Figure 3: Weak acid and strong base titration curve

The equivalence point for this case **lies above the neutral pH**, indicating a basic state at equal parts weak acid and strong base. This is because once all H^+ is neutralized by the presence of OH^- from NaOH, **what is left is the strong conjugate base molecule** CH_3COO^-. In the presence of this strong conjugate base, some water molecules will react with the CH_3COO^- to form CH_3COOH and OH^-, thus resulting in a higher pH value at equivalence. When confused, it's helpful to (1) assume the reaction will go to completion and then (2) look at what is left and ask how it affects the pH.

Core Concept 3.4: Buffers

Buffers are solutions containing both a weak acid and its corresponding conjugate base. They are used to **resist** large changes of pH upon addition of strong acids or bases.

Based on the chemical equation of a weak acid ($HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$), we can manipulate the K_a equation to give us the final following formula.

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = [H_3O^+]\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

$$-\log K_a = -\log[H_3O^+] + \log\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

$$pH = pK_a + \log\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

The final equation is a simple way to determine the pH of a buffer solution under a given concentration of the weak acid and its conjugate base. *Notice how the pH of a solution will be equivalent to the pK_a when there are equal parts conjugate acid and base*

The making a buffer solution can be done in one of three ways.

(a) Equal parts of the weak acid and its conjugate base

(b) Weak acid with half the amount of a strong base

(c) Conjugate base with half the amount of a strong acid

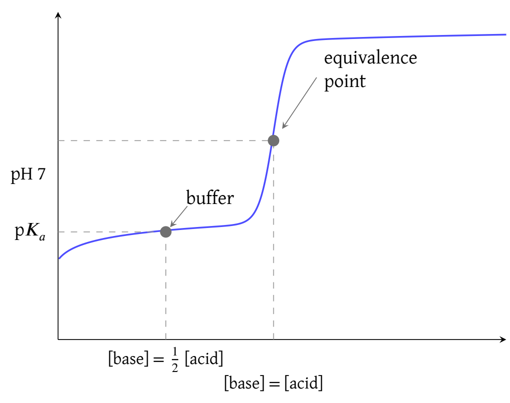

Figure 4: Buffer solution titration curve

Based on Figure 4 the pK_a is also half way between the equivalence point and y-axis of the titration curve, the point at which [conjugate acid] = [conjugate base]. Any subsequent addition of a strong acid or base will shift the pH value, as shown in the following titration curve, but the pH will be resistant to change within a given range until larger amounts of a strong acid or base is added.

Using this technique, chemists can customize the pH they want a solution to linger around based on the K_a value of a weak acid. Conversely, a buffering solution can also be held at a basic pH level by adding equal parts of weak base with its corresponding strong conjugate acid.